Erkan Tairi

Postdoctoral Researcher, UC Berkeley
Research Scientist, Alloc Init Labs

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Research Interests

Foundations of cryptography, public-key cryptography, applied cryptography

Experience

Oct 2025 – present	**Postdoctoral Researcher, UC Berkeley** Host: Sanjam Garg	Berkeley, CA, USA
Sep 2025 – present	**Research Scientist, Alloc Init Labs** Cryptographic protocol design and documentation	Remote
May 2024 – Sep 2025	**Postdoctoral Researcher, ENS Paris** Host: David Pointcheval	Paris, France
Feb 2022 – Jan 2024	**Research Engineer, MyPrivacy** Cryptographic protocol design and development	Vienna, Austria

Education

2019 – 2024	**PhD in Computer Science, TU Wien** Supervisors: Matteo Maffei (TU Wien) and Daniel Slamanig (AIT Austrian Institute of Technology) Thesis: *Foundations of Adaptor Signatures for Distributed Ledger Protocols* DOI	Vienna, Austria
2016 – 2018	**MSc in Computer Science, Johannes Kepler University**	Linz, Austria
2011 – 2015	**BSc in Computer Science, University St. Paul the Apostle**	Ohrid, Macedonia

Awards and Grants

Fellowship	Erwin Schrödinger Fellowship, Austrian Science Fund (FWF) DOI
Grants	France-Berkeley Fund Input | Output Research Grant Internet Freedom Fund, Open Technology Fund

Publications

Published

IACR CiC to appear
ePrint

Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded Depth Circuits and Turing Machines, and More
Tapas Pal, Robert Schädlich and **Erkan Tairi**

USENIX Security 2026
ePrint Proceedings

Lighthouse: Single-Server Secure Aggregation with O(1) Server-Committee Communication at Scale
Sanjam Garg, Alireza Kavousi, Dimitris Kolonelos, **Erkan Tairi** and Zhipeng Wang

CRYPTO 2026
ePrint DOI

LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via 2HashDH Paradigm

Muhammed F. Esgin, Ron Steinfeld, **Erkan Tairi** and Jie Xu

SCN 2026
ePrint DOI

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE

Valerio Cini and **Erkan Tairi**

SCN 2026
ePrint DOI

Ciphertext-Updatable Attribute-Based and Predicate Encryption from Lattices

Robert Schädlich, Linda Scheu-Hachtel, **Erkan Tairi** and Yuejun Wang

PKC 2026
ePrint DOI

Hardness of M-LWE with General Distributions and Applications to Leaky Variants

Katharina Boudgoust, Corentin Jeudy, **Erkan Tairi** and Weiqiang Wen

PKC 2026
ePrint DOI

On Verifiable Delay Functions from Time-Lock Puzzles

Hamza Abusalah, Karen Azari, Dario Fiore, Chethan Kamath and **Erkan Tairi**

EUROCRYPT 2026
ePrint DOI

Impossibility of VDFs in the ROM: The Complete Picture

Hamza Abusalah, Karen Azari, Chethan Kamath, **Erkan Tairi** and Maximilian von Consbruch

EUROCRYPT 2024
ePrint DOI

Lower Bounds for Lattice-based Compact Functional Encryption

Erkan Tairi and Akin Ünal

Journal of Cryptology
ePrint DOI

(Inner-Product) Functional Encryption with Updatable Ciphertexts

Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and **Erkan Tairi**

ACM CCSW 2023
DOI

Optimizing 0-RTT Key Exchange with Full Forward Security

Christian Göth, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks, **Erkan Tairi** and Alexander Zikulnig

ACM CCS 2023
ePrint DOI

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures

Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind

ACM CCS 2022
ePrint DOI

Foundations of Coin Mixing Services

Noemi Glaeser, Matteo Maffei, Giulio Malavolta, Pedro Moreno-Sanchez, **Erkan Tairi** and Sri AravindaKrishnan Thyagarajan

IEEE S&P 2021
ePrint DOI

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs

Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei

FC 2021
ePrint DOI

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments

Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei

PKC 2021
ePrint DOI

Updatable Signatures and Message Authentication Codes

Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and **Erkan Tairi**

Preprints (Under Submission)

2026
ePrint

Gaussian Kernel Lattices and Smoothing Bounds from Theta Integrals

Samed Düzlü, Nihar Gargava and **Erkan Tairi**

Algebraic Cryptanalysis of Hinted vSIS and Decomposed LWE

Valerio Cini, Robert Schädlich and **Erkan Tairi**

Lower Bounds for Linear Secret Sharing with Bounded Reconstruction Coefficients

Yiping Ma and **Erkan Tairi**

ePrint

Decentralized Multi-Authority (Attribute-Based) Traitor Tracing

Pratish Datta, Robert Schädlich and **Erkan Tairi**

ePrint

Adaptor Signatures Meet BLS: Enabling Efficient Blockchain Applications with Unique Adaptor Signatures

Javier Gomez-Martinez, **Erkan Tairi**, Pedro Moreno-Sanchez and Clara Schneidewind

ePrint

Foundations of Verifiably Encrypted (Blind) Signatures
Diego Castejon-Molina, **Erkan Tairi**, Dimitrios Vasilopoulos and Pedro Moreno-Sanchez

Selected Talks

2026

Lighthouse: Single-Server Secure Aggregation with O(1) Server-Committee Communication
Privacy-Preserving Machine Learning Workshop, Aug 2026
Bay Area Crypto Day, May 2026

LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via 2HashDH Paradigm
CRYPTO 2026, Aug 2026 Video

2025

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE
Crypto Day at Télécom Paris, Jun 2025 shorturl.at/DrJEr
Guest talk at King's College London Cybersecurity Group, Feb 2025

2023

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures
IOG Seminar, Nov 2023

2022

(Inner-Product) Functional Encryption with Updatable Ciphertexts
PICOCRYPT Seminar at IMDEA Software Institute, Dec 2022 shorturl.at/5RP3o

2021

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs
IEEE Symposium on Security and Privacy, May 2021 Video

Updatable Signatures and Message Authentication Codes
PKC 2021, May 2021 Video
Young Researcher Crypto Seminar, May 2021

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments
Financial Cryptography and Data Security 2021, Mar 2021 Video
Decrypto Seminar, Dec 2020

Service

Program Committee
USENIX Security 2027, ACM CCS 2025–2026, ACM AsiaCCS 2026, FC 2025–2026, IACR Communications in Cryptology 2025–2026, CVC 2025

External Reviewer
ACISP 2024; ACM CCS 2021–2023; ACM AFT 2022; ACNS 2024–2025; APKC 2021–2022; Asiacrypt 2021, 2024–2025; CANS 2022; Crypto 2023–2026; Eurocrypt 2023, 2025–2026; FC 2021–2024; IEEE S&P 2024; IWSEC 2021–2023; PKC 2026–2027; ProvSec 2020–2023

Organization
Co-organizer of the ViSP Cryptography Research Meetup
Member of the NDSS 2023 Student Support Committee

Teaching and Supervision

2019 – 2023

Teaching Assistant, TU Wien Vienna, Austria
Introduction to Cryptography, Privacy-Enhancing Cryptography, Cryptocurrencies (graduate level)

MSc Theses
Aron Wussler (TU Wien): *Post-quantum Cryptography in OpenPGP* DOI
Ngaffo Raissa Flora (University of Bamenda): *Simulation-extractable NIZK Proof Systems*

Research Visits and Internships

Oct 2026	**Tsinghua University** Host: Yang Yu	Beijing, China
May 2026	**ENS Paris** Host: Brice Minaud	Paris, France
Feb 2025	**King's College London** Host: Martin Albrecht	London, England
Dec 2022	**IMDEA Software Institute** Host: Dario Fiore	Madrid, Spain
Nov 2022	**ETH Zürich** Host: Dennis Hofheinz	Zürich, Switzerland
Oct 2018 – Mar 2019	**Intern, TU Wien** Supervisor: Matteo Maffei	Vienna, Austria
Jun – Sep 2018	**Intern, AIT Austrian Institute of Technology** Supervisor: Daniel Slamanig	Vienna, Austria

Skills

Programming	C, C++, Rust (*advanced*); C#, Python, Shell (*average*); Java, Go (*basics*)
Web	HTML, CSS, JavaScript (React, Node.js), ASP.NET Web (*advanced*)
Software	Visual Studio, SageMath, Magma, Git, LaTeX
Other	Microsoft Azure, Amazon AWS (cloud computing)

Languages

Native	Macedonian
Fluent	English, Turkish (*professional working proficiency*)
Intermediate	French, German, Albanian (*elementary proficiency*)

References

Sanjam Garg
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David Pointcheval
david.pointcheval@ens.fr

Daniel Slamanig
daniel.slamanig@unibw.de

Matteo Maffei
matteo.maffei@tuwien.ac.at